

Electronics Design Project

BJT Amplifier

Carissa Lee

UVM Department of Electrical Engineering

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Instructor: Dr. James Kay

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Abstract

This project involved the design, numerical analysis, simulation, and experimental validation of a transistor-based analog voltage amplifier circuit. The system was required to amplify an input alternating current signal by a factor of greater than 250V/V, with an output swing of a minimum of 5V peak to peak, and a mid band gain between 100Hz and 200kHz. Other constraints applied to the design such as power consumption, bias stability, source resistance, output resistance, and compliance with small signal model parameters.

Results were examined through a test plan constructed from user-based engineering requirements that set a foundation for the design process. The outcome demonstrated reliable voltage amplification of 497V/V, an output swing of greater than 6.4Vpp and a wide midband from 94Hz to 251kHz. Simulation confirmed experimental results and numerical analysis, verifying the design and implementation of the device.

Many trade-off style constraints were identified, such as gain vs. bandwidth and stability vs. gain, and were explored to identify what parameters allowed for the meeting of all criteria. Bias stability was analyzed for sensitivity to device parameters such as DC current gain (β) and early voltage affects (V_A). Future work should include the implementation of different circuit topologies that may give rise to larger gain, wider bandwidth, greater bias stability, or the need for less transistors used. Overall, this design meets all function requirements and provides a robust voltage amplifier.

Requirements

The purpose of this system is to amplify an alternating current input signal through an analog circuit. To ensure reliability, customer satisfaction, and consistency in performance, a set of engineering requirements were established to define functional behavior restrictions that govern the operation of the active circuit. These requirements strongly influenced design, and were translated into a test plan for downstream verification and validation.

Design Requirements

- 1000: The amplifier shall have a small-signal output resistance of less than 50 Ω measured at the output node under midband operating conditions.
- 1010: The device shall provide a midband small-signal voltage gain with magnitude greater than 250 between the input and overall output terminals of the circuit when driven by a source with an output resistance of 10 k Ω .
- 1020: The amplifier's lower cutoff frequency shall be below 100 Hz, where this cutoff is defined as the frequency at which the gain magnitude decreases by 3 dB from the midband value.
- 1030: The amplifier's higher cutoff frequency shall be above 200 kHz, where this cutoff is defined as the frequency at which the gain magnitude decreases by 3 dB from the midband value.
- 1040: The device shall operate from a single 15 V DC power supply.
- 1050: The quiescent collector current of each transistor shall vary by no more than 6% when any transistor is replaced by another device of the same type, accounting for Early voltage and DC current gain variations.

- 1060: All performance specifications shall be met when the amplifier is driven by a source with a resistance of $10\text{ k}\Omega$.
- 1070: The device shall produce an undistorted output voltage swing of at least 5 V peak-to-peak at the output node while remaining in the forward-active region of operation.
- 1080: The device shall have a total quiescent power consumption of less than 150 mW across all transistors.
- 1090: The device shall use no more than four transistors in the completed design.
- 1100: For each common-emitter stage, the design shall have a $g_m R_C$ product of less than 250.

Results

The device was tested in accordance with the engineering requirements, following a defined verification plan, this full plan and proof of achieving all requirements can be found in Appendix A. Table 1 below summarizes requirement level verification, while Table 2 provides quantitative performance metrics. All engineering requirement were achieved within specified tolerances.

Table 1: Requirements Verification Summary

Test ID	Requirement	Measured Outcome	Result
0100	Output resistance in Midband $< 50 \Omega$	R_{out} verified via simulation and analysis	PASS
0110	Voltage gain > 250 with $R_s = 10 \text{ k}\Omega$	Gain verified on oscilloscope and in SPICE	PASS
0120	Bandwidth from 100Hz to 200kHz	Measured midband surpasses	PASS
0130	Operates from a single 15 V supply	Correct operation at $V_{CC} = 15 \text{ V}$	PASS
0140	Bias variation ΔI_C is $< 6\%$	Confirmed $< 6\%$ variation	PASS
0150	Output swing of $> 5 \text{ V}_{pp}$	Output swing greater than 5V	PASS
0160	Quiescent power $< 150 \text{ mW}$	Confirmed $< 150 \text{ mW}$	PASS
0170	Uses no more than four transistors	Uses four BJTs	PASS
0180	CE stages satisfies $g_m R_C < 250$	Confirmed $g_m R_C < 250$	PASS

Table 2: Performance Ranges and Sensitivities

Metric	Analytical	Experimental	Notes
Midband gain	630 V/V	497 V/V	No clipping
Lower cutoff frequency f_L	$< 10 \text{ Hz}$	94 Hz	From 3 dB cutoff
Upper cutoff frequency f_H	344.4 kHz	251.2 kHz	From 3 dB cutoff
Midband range	$< 10 \text{ Hz}$ to 344.3 kHz	94 Hz to 251.2 kHz	Greater than requirement
Output resistance R_{out}	12.9 Ω	26.2 Ω	Below requirement
Quiescent power P_Q	72.6 mW	66.84 mW	Much below requirement
Output swing	$\approx 8.1 \text{ V}_{pp}$	$> 5 \text{ V}_{pp}$	1 kHz input
Bias sensitivity to β , V_A	—	$\Delta I_C / I_C < 6\%$	SPICE performed
Hybrid-pi validation	$g_m R_C = 225.54$	—	—

As summarized in Table 1, all defined engineering requirements were satisfied from numerical, analytical, or experimental validation. The device reliably amplified an AC input signal while providing a mid-band gain of greater than 250. The frequency bandwidth measured from the -3dB points showed a bandwidth greater than the required 100Hz to 200kHz while output swing was greater than 5V_{pp}, demonstrating reliable signal amplification across a wide range of inputs. AC

model parameters were validated, such as ensuring $g_m R_C < 250$ with a low output resistance. Bias sensitivity across transistor variability, DC current gain and early voltage effects, proved minimal with collector current of each transistor varying less than 6%.

Design and Analysis

Block Diagram

At a system level, the operation of the circuit can be decomposed into a series of functional blocks that work to take an input signal and produce an amplified output. Figure 1 below shows the block diagram used to guide the design process.

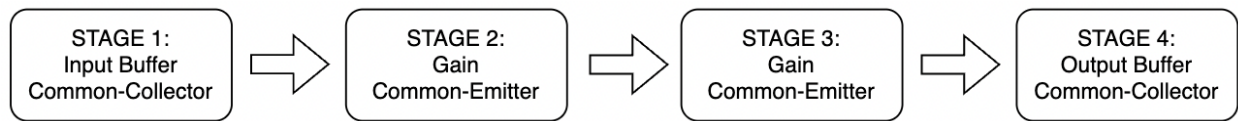


Figure 1: Functional Block Diagram of BJT Amplifier

To begin the series, stage one implements a common-collector topology to act as an input buffer. The following two stages create a cascaded common-emitter section that serves to amplify the input. Finally, stage four is a common-collector that serves as an output buffer.

Circuit Topology

With the basic system architecture being defined at a block diagram level, the circuit implementation is shown in Figure 2 below, demonstrating outlined functional behavior.

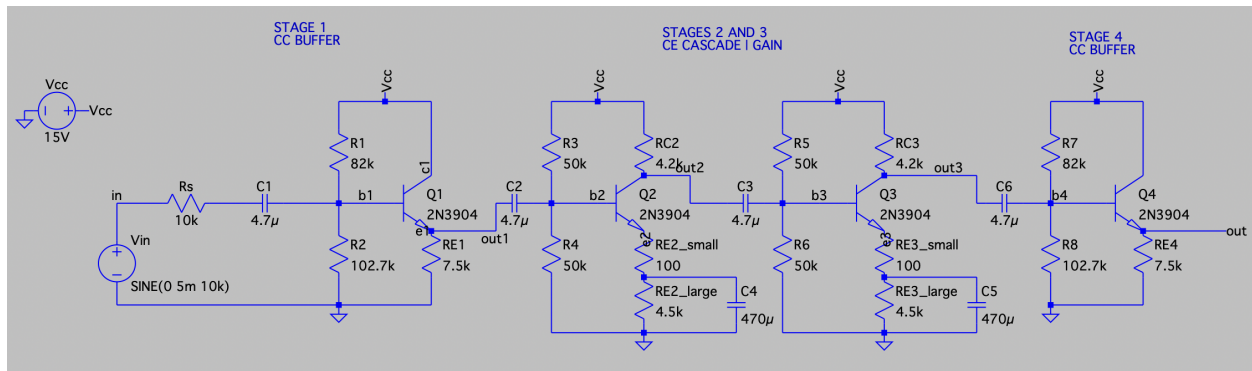


Figure 2: Full Analog Schematic of Four Stage BJT Amplifier

The created experimental circuit is shown below in Figure 3.

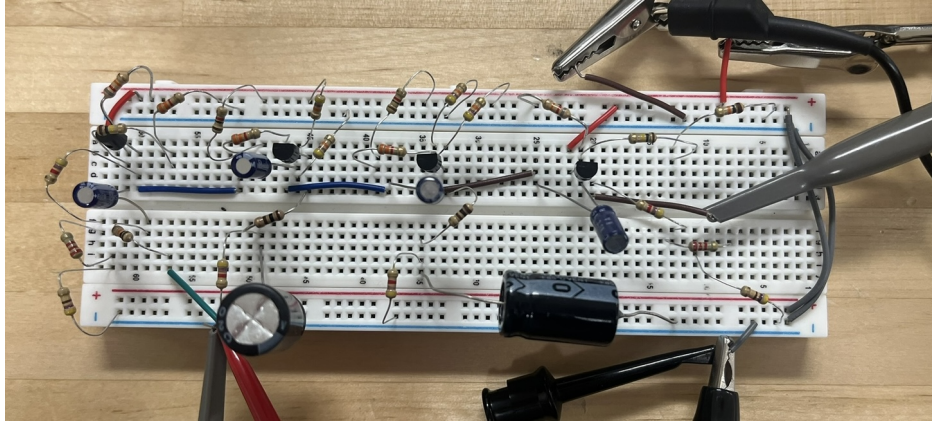


Figure 3: Experimental Circuit BJT Amplifier

To better understand the emergence of the systems behavior, the subsections of the circuit can be broken down and analyzed to better understand each separate section.

4.2.1 Input Buffer Common-Collector

Referring back to the block diagram, the first section describes an input buffer stage that separates the source voltage and resistance from the gain stages. This was created using analog components, realized below in Figure 4.

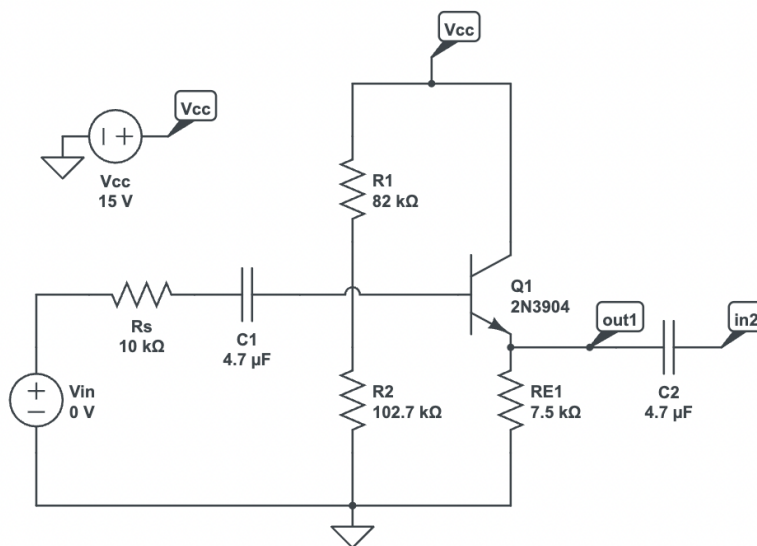


Figure 4: Stage 1: Input Buffer Functional Block Circuit Schematic

This buffer stage is designed the same as Stage 4, allowing for symmetrical analysis. The transistor used for this analysis is the NPN 2N3904 small signal switching bipolar junction transistor. It's DC current gain can fluctuate from 80 to 300, and for the DC analysis, a quiescent

$\beta = 100$ is chosen. Referring back to Figure 4, this stage is in a common-collector or emitter-follower design, and it acts as an input buffer between the source and the high-gain amplification stages. The source resistance (R_S) is $10k\Omega$, as specified within the requirements. This buffer stage is required as later gain stages (stages 2 and 3) present a low input resistance which would introduce significant loading on the source. Had the source been directly coupled to the gain stages, the source resistance and amplifier input resistance would form a voltage divider, reducing the effective input signal, making it more difficult to reach higher gain values. The emitter follower provides a high input resistance, a low output resistance to strongly drive the following stage, and a unity gain to preserve signal amplitude.

The DC bias point at the output of transistor one is first chosen through a voltage divider between the two input resistors R_1 and R_2 . This value was chosen to bias the emitter output near mid-supply to allow for large emitter swing without clipping. The DC emitter voltage was hence chosen to sit at half of the power supply ($7.5V$), hence the base voltage was chosen to sit at approximately V_{BEon} ($0.7V$) above this value. The exact resistor and bias values are calculated using Equation 1 below.

$$V_B \approx V_{CC} \frac{R_2}{R_1 + R_2} : V_B \approx 15 \frac{102.7k}{82k + 102.7k} \approx 8.34V \quad (1)$$

This means that the emitter DC voltage is held V_{BEon} below this, shown below in Equation 2.

$$V_E \approx V_B - V_{BEon} \approx 8.34V - 0.7V = 7.64V \quad (2)$$

The output for this stage is taken at the emitter node, which is held at $7.64V$, roughly at the mid supply allowing for a wide voltage swing with conservative headroom.

The emitter current is then chosen to be $1.00mA$ to provide a reasonably low output resistance while keep quiescent power low. The emitter resistor can then be calculated using Ohms law, shown below in Equation 3.

$$R_E = \frac{V_E}{I_E} : \frac{7.64V}{1.00mA} = 7.5k\Omega \quad (3)$$

The bias resistors R_1 and R_2 were also chosen to be "stiff", behaving as a strong voltage source with low thevenin resistance, to decrease sensitivity to DC current gain (β) fluctuations. This is done by making the divider current significantly larger than the base current. The divider current is represented below in Equation 4.

$$I_{div} = \frac{V_{CC}}{R_1 + R_2} : \frac{15V}{184.7k\Omega} \approx 81\mu A \quad (4)$$

The base current can then be calculated as a function of the collector current.

$$I_B = \frac{I_C}{\beta} : \frac{1.00mA}{100} = 10\mu A \quad (5)$$

This means that the divider is 8x larger than the base current, reducing bias drift when swapping transistor parameters. These quiescent bias circuit parameters are summarized below in Table 3.

Table 3: Quiescent Operating Point Summary for Stage 1 (Common-Collector Buffer)

Quantity	Symbol	Value	Units
Supply Voltage	V_{CC}	15	V
Collector Voltage	V_C	15	V
Base Voltage	V_B	8.34	V
Emitter Voltage	V_E	7.64	V
Collector-Emitter Voltage	V_{CE}	7.36	V
Emitter Current	I_E	1.00	mA
Collector Current	I_C	1.00	mA
Base Current	I_B	10	μA
Divider Current	I_{div}	81	μA
Emitter Resistor	R_E	7.5	k Ω
Bias Resistors	R_1, R_2	82, 102.7	k Ω

The coupling capacitors C1 and C2 create an effective high pass filter with the source resistor and the input resistance. Given that the bandwidth needs to include 100Hz, the lower cutoff frequency must be very low. Equation 6 below shows the effective cutoff frequency.

$$f_{C1} = \frac{1}{2\pi C_1(R_S + R_{in})} : \frac{1}{2\pi(4.7\mu F)(55k\Omega)} \approx 0.6Hz. \quad (6)$$

The choice of a 4.7uF coupling capacitor provides a very low cutoff frequency of 0.6Hz, not threatening the lower bandwidth frequency requirement. The addition of a coupling capacitor between each stage also allows for independent analysis and biasing.

The upper cutoff frequency can then be calculated using Equation 7 below which uses the datasheet-specified parasitic capacitance that set the low-pass corner cutoff frequency.

$$C_{in} \approx C_\pi + (1 - A_V)C_\mu \quad (7)$$

Where C_π is the base-emitter capacitance, C_μ is the base-collector capacitance, and A_V is the voltage gain of the stage.

The equivalent resistance seen by this capacitance is the parallel combination of the source resistance and the input resistance,

$$R_{eq} = R_S || R_{in} \quad (8)$$

The upper cutoff frequency can then be estimated using the first-order RC pole relationship,

$$f_H \approx \frac{1}{2\pi(R_S || R_{in})C_{in}} : \frac{1}{2\pi(10k\Omega || 43k\Omega)(26pF)} \approx 760kHz \quad (9)$$

This is expected as the emitter-follower topology offers a wide bandwidth and will not serve as the limiting factor of this device.

Mid-band impedance behavior was then considered to ensure that this stage presents high input resistance and low output resistance to drive the following stage. The small signal parameter values for transconductance, small-signal emitter resistance, and base-emitter small-signal resistance are calculated below in Equations 10, 11, and 12 respectively.

$$g_m = \frac{I_C}{V_T} : \frac{1.00mA}{25.9mV} = 0.039S \quad (10)$$

$$r_e = \frac{1}{g_m} : \frac{1}{0.039S} = 25.6\Omega \quad (11)$$

$$r_\pi = \frac{\beta}{g_m} : \frac{100}{0.039S} = 2.56k\Omega \quad (12)$$

From these values, the input resistance into the base can be found in Equation 13 below.

$$R_{inbase} \approx r_\pi + (\beta + 1)R_E : 2.56k\Omega + 101(7.5k\Omega) \approx 760k\Omega \quad (13)$$

The bias divider loads the input in parallel, shown below in Equation 14.

$$R_B = R_1 || R_2 : 82k\Omega || 102.7k\Omega \approx 45.6k\Omega \quad (14)$$

The input resistance can then be found as the parallel combination of the bias resistors and the input base resistance, seen below in Equation 15

$$R_{in} = R_1 || R_2 || R_{inbase} : 82k\Omega || 102.7k\Omega || 760k\Omega = 43k\Omega \quad (15)$$

Since the input resistance is significantly greater than the source resistance, the source is not heavily loaded, meaning that the buffer preserves most of the source signal while isolating gain.

The output resistance can then be found as the parallel combination of the quotient of the thermal voltage and quiescent collector current, the emitter resistor, and the resistance from the early affect. This is below in Equation 16.

$$R_o = \frac{V_T}{I_{CQ}} || R_E || r_o \quad (16)$$

Given that the early voltage effect is set to be 100V, the value of its resistance is found as the early affect voltage over the quiescent collector current, realized below in Equation 17

$$r_o \approx \frac{V_A}{I_{CQ}} : \frac{100V}{1mA} \approx 100k\Omega \quad (17)$$

Substituting this into Equation 16, the output voltage is found to be 25.9Ω.

Finally, this stage's voltage gain can then be derived from the topology, shown below in Equation 18.

$$A_V = \frac{(1 + \beta)(r_o || R_E)}{r_\pi + (1 + \beta)(r_o || R_E)} \frac{R_{in}}{R_{in} + R_s} \quad (18)$$

From this, the voltage gain of this stage is calculated to be 0.81, a value expected for this topology. Table 4 below summarizes the hybrid-pi model values for Stage 1 below.

Table 4: Small-Signal Midband Parameter Summary for Stage 1 (Common-Collector Buffer)

Quantity	Symbol	Value	Units
Transconductance	g_m	0.039	S
Intrinsic emitter resistance	r_e	25.6	Ω
Base-emitter resistance	r_π	2.56	k Ω
Base bias resistance	R_B	45.6	k Ω
Input resistance into base	$R_{in,base}$	760	k Ω
Total input resistance	R_{in}	43	k Ω
Early-effect resistance	r_o	100	k Ω
Output resistance	R_{out}	25.9	Ω
Midband voltage gain	A_V	0.81	V/V

Overall, the common-collector input buffer meets its intended roles as an isolating stage. The calculated input resistance (43k Ω) is sufficiently larger than the source resistance, preventing input signal loading. Alongside this, the predicted output resistance (25.9 Ω) is low enough to drive the following common-emitter stage. The mid-band gain (0.81V/V) is consistent with the expected value, and the surrounding coupling capacitors (4.7 μ F) places the input high-pass corner well below the 100Hz requirement. Alongside this, the DC operating point is biased near mid-supply to allow adequate headroom for symmetrical output swinging without clipping near the rails (15V and 0V).

4.2.2 Gain Common-Emitter Cascade

Stages two and three are designed as a common-emitter cascade for voltage gain. Each stage is designed to have the same components and produce the same midband gain while maintaining stable biasing and adequate bandwidth. The design for these two stages are shown below in Figure 5.

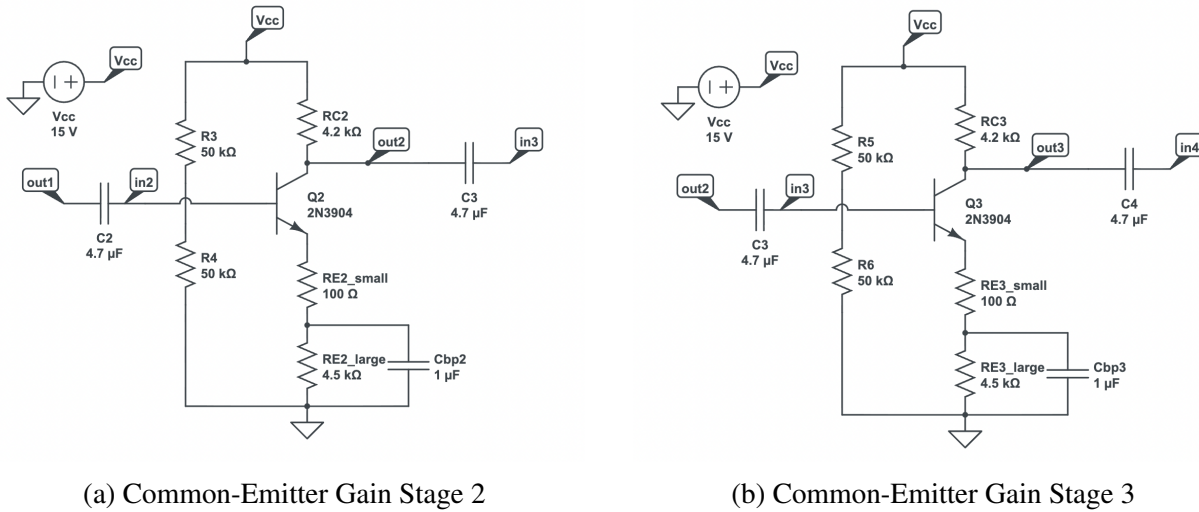


Figure 5: Design of cascaded common-emitter voltage gain stages (Stages 2 and 3).

This analysis is the same for Stages 2 and 3 as they were designed the same. The gain was split across two stages to reduce the required per-stage gain to improve on bandwidth while limiting the Miller effect in any singular transistor. This also allowed for the production of a higher gain than allowed by a single-stage gain amplifier.

To begin, the bias resistor network between R3 and R4 are chosen to set the base at mid supply to allow the collector swing to be symmetrical. Setting the two parallel bias resistors equal to one another would center the base. These specific values were chosen to make the divider current much larger than the base current, for bias stability, while not wasting too much power. For this, $R_3 = R_4 = 50k\Omega$. The calculated centered base voltage is shown below in Equation 19.

$$V_B = \frac{R_4}{R_4 + R_3} V_{CC} : V_B = \frac{50k}{100k} 15 = 7.5V \quad (19)$$

The emitter voltage is then found to be approximately $0.7V$ (V_{BEon}) below the base voltage, giving $V_E \approx 6.8V$. The emitter current is calculated through Ohm's law, yielding $I_E \approx 1.48mA$. The collector current is then related to the emitter current below in Equation 20.

$$I_C = I_E \frac{\beta}{\beta + 1} \quad (20)$$

Using $\beta \approx 100$,

$$I_C = (1.48mA) \frac{100}{101} \approx 1.46mA. \quad (21)$$

To select the collector resistor, KVL is applied around the collector loop:

$$V_C = V_{CC} - I_C R_C. \quad (22)$$

Rearranging to solve for R_C gives

$$R_C = \frac{V_{CC} - V_C}{I_C}. \quad (23)$$

Choosing $R_C = 4.2k\Omega$ yields the quiescent collector voltage

$$V_C = 15V - (1.46mA)(4.2k\Omega) = 8.87V. \quad (24)$$

This value is near mid supply, making it a good bias point for voltage swing symmetry. Table 5 below summarizes the quiescent operating points for Stages 2 and 3.

Table 5: DC Operating Point for Common-Emitter Gain Stage

Quantity	Symbol	Value	Units
Supply Voltage	V_{CC}	15	V
Collector Voltage	V_C	10.8	V
Base Voltage	V_B	8.34	V
Emitter Voltage	V_E	7.64	V
Collector-Emitter Voltage	V_{CE}	3.16	V
Emitter Current	I_E	1.00	mA
Collector Current	I_C	1.00	mA
Base Current	I_B	10	μA
Divider Current	I_{div}	81	μA
Emitter Resistor	R_E	7.5	k Ω
Collector Resistor	R_C	4.2	k Ω
Bias Resistors	R_1, R_2	82, 102.7	k Ω

Following this, the small signal AC parameters are analyzed to ensure the gain stage is designed as intended.

The small signal emitter resistance can be found as the quotient of the thermal voltage and emitter current, found to be 16.9 Ω . This value can then be used to find the effective mid-band gain below in Equation 30.

$$A_v = \frac{-R_c}{r_e + 100} : \frac{-4200}{16.9 + 100} = -35.9 \quad (25)$$

So, the magnitude gain of each stage gain is 35.9V/V.

Following Equations 7-11, the rest of the small-signal parameters can be found for Stages 2 and 3, and are summarized below in Table 6.

Table 6: Small-Signal Parameters for Common-Emitter Gain Stage

Quantity	Symbol	Value	Units
Transconductance	g_m	0.0584	S
Intrinsic emitter resistance	r_e	17.1	Ω
Base-emitter resistance	r_π	1.71	k Ω
Base bias resistance	R_B	25.0	k Ω
Input resistance into base	$R_{in,base}$	11.8	k Ω
Total input resistance	R_{in}	8.02	k Ω
Early-effect resistance	r_o	100	k Ω
Output resistance	R_{out}	4.03	k Ω
Mid-band voltage gain	A_v	-34.4	V/V

Finally, the frequency content was taken into account when choosing the coupling and bypass capacitors. Similar to Stage 1, the coupling capacitors were all held at 4.7 μF to maintain a low high-pass corner cutoff frequency, as previously calculated. The emitter bypass capacitors were chosen to form a low-frequency high pass corner so the emitter resistors can provide strong DC bias

stability while removing the AC signal across the reject-band. This is shown below in Equation 26.

$$f_{Cb2} = \frac{1}{2\pi R_{E2,large} C_{bp2}} : \frac{1}{2\pi(4.5k\Omega)(1\mu F)} \approx 35.4Hz \quad (26)$$

The lower cutoff frequency set by the bypass capacitor is 35.4Hz. The higher cutoff frequency can also be determined, similarly to from the input buffer stage, through analyzing the Miller effect from junction and parasitic capacitance. This is calculated below in Equation 27.

$$f_{H,M} \approx \frac{1}{2\pi(R_{S,eq} || R_{in})(C_{\pi} + (1 - A_V)C_{\mu})} \quad (27)$$

The upper cutoff low-pass frequency is calculated to be 362kHz, above the required value. Given that this value is lower than the determined cutoff of the buffer stages, the entire circuits bandwidth will be set with these values as they are the limiting factor.

The total numerical gain can be found as the product of each individual stage, this is calculated to be 776 V/V below in Equation 28.

$$A_v \approx (34.3)(34.4)(0.81)(0.81) = 776 \quad (28)$$

This value is conservatively above the required gain value, exceeding this requirement.

4.2.3 Output Buffer Common-Collector

The final stage has the same design as Stage 1, forming a common-collect (emitter-follower) topology. It's purpose is not to provide any additional voltage gain, as it actually removes gain due to its value of 0.81, but rather to reduce overall output resistance of the amplifier. The emitter-follower provides a high input resistance and low output resistance, making it so the prior stage is not loaded and the full signal is delivered to the load. The biasing mirrors that on the input buffer, positioning the emitter output near mid-supply to allow for adequate voltage swing without signal clipping/distortion. The stage improves drive capability and preserves signal integrity.

Theory of Operation

After analyzing the design of each functional group, the emergence of the circuit behavior can be discussed in an idle and active state to gain insight into how the system works.

4.3.1 Idle State

Under idle conditions, the input signal to the amplifier is ideally zero, containing only background noise that is often negligible. Because each stage is AC coupled through coupling capacitors, no DC signal from the source propagates, rather, each transistor remains biased at its designed quiescent operating points. Various voltage divider networks establish base voltages, which set stable emitter and collector currents while keeping power consumption low. Due to this, each transistor operates in the forward active region with a constant collector current and voltage.

Without significant alternating current input, the small-signal variation is roughly zero. This makes the output of each stage remaining at its DC biased level, with the final output remaining centered around its quiescent emitter voltage. The amplifier remains in a linear, stable state with no large signal delivered to the load.

4.3.2 Event State

When an AC input is applied, it becomes added onto the DC bias input of the first stage. The input buffer presents a high input impedance to prevent source loading, and passes the AC signal to the gain stage without notable attenuation.

In the common-emitter cascade gain stages, the small AC input produces a proportional variation in collector current, resulting in an amplifier voltage swing across the output. Because these stages are designed to operate in a linear, forward active mode, with sufficient headroom, the signal is amplified without device saturation. The bypass emitter resistor allows for significant voltage gain while allowing for DC bias stability against DC current gain and early voltage effects.

The amplifier signal is then passed to the final buffer (emitter follower) stage which serves to lower the output resistance to account for the engineering requirement. This allows for the amplifier to drive the load without altering the signal. Hence, the output is a clean, non-distorted, scaled version of the original input signal, amplified by the designed mid-band gain.

Bias analysis

To begin the LTSpice analysis, a DC operating point simulation was conducted to verify that the biasing points across the four stages were as intended under quiescent conditions. This is very important to ensure a full, symmetrical output swing and to maintain linear operation of each transistor.

For the DC analysis, the input was set to 0V DC to imitate the lack of an input signal to be amplified, in which the output would ideally be held around mid-rail. Under direct current operation, all capacitors act as opens, blocking steady state voltage between stages to allow downstream operating points to be analyzed independent of the input stage bias. Figure 6 below shows the LTSpice circuit schematic.

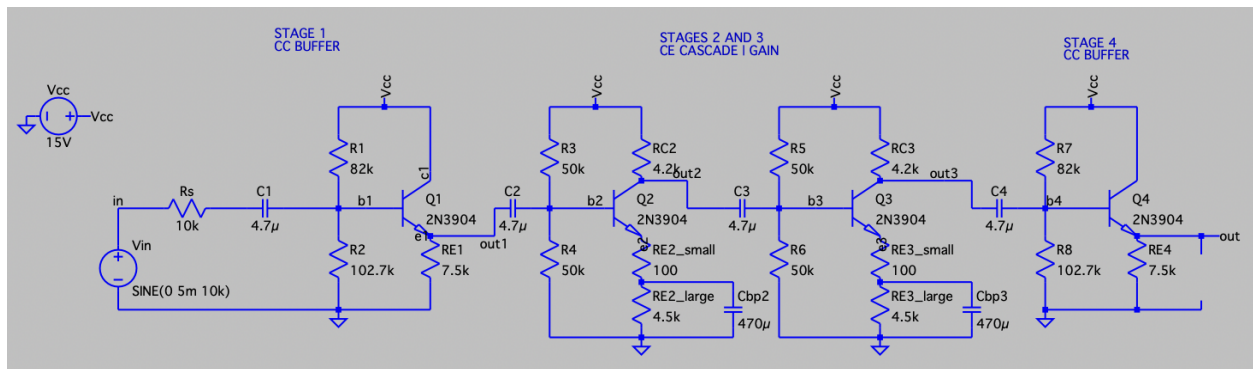


Figure 6: DC Analysis LTSpice Circuit Schematic

The DC operating points were evaluated at the following critical nodes: the base, collector, and emitter voltages of each transistor. Figure 7 below shows the values recorded at each of these nodes for the emitter-followers (left) and the common-emitters (right).

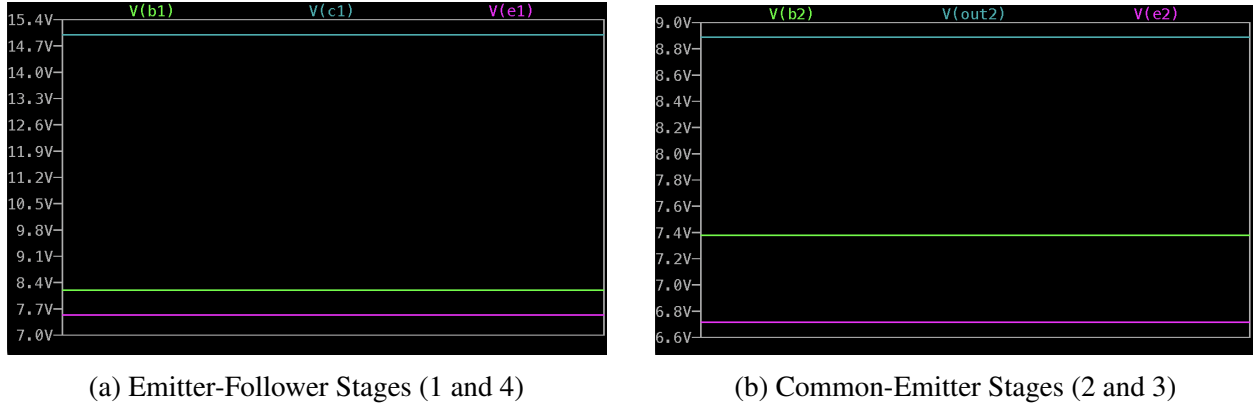


Figure 7: DC operating point values recorded for each amplifier topology. Stages 1 and 4 share identical emitter-follower configurations, while Stages 2 and 3 share identical common-emitter configurations; therefore, each pair requires only one analytical treatment.

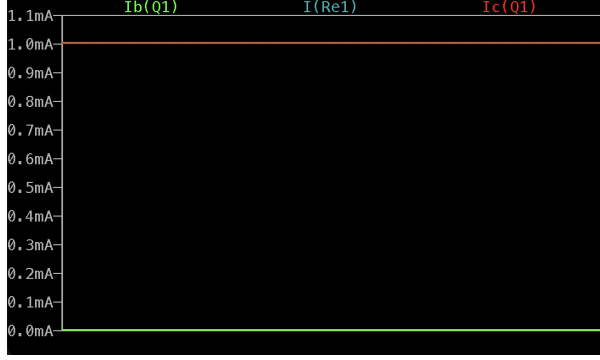
Table 7 below summarizes the DC values found at each of these nodes.

Table 7: DC Node Voltages for Each Amplifier Stage

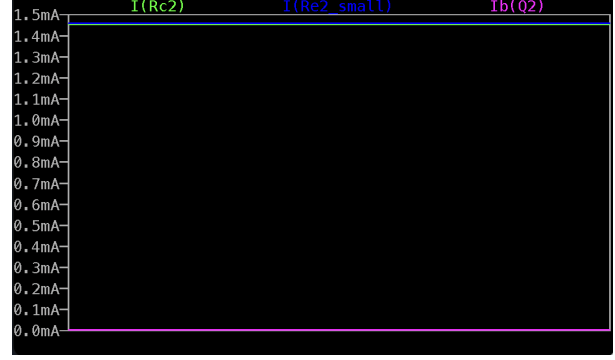
Stage(s)	Topology	V_B (V)	V_C (V)	V_E (V)	V_{CE} (V)
1 and 4	Common-Collector	8.34	15.0	7.64	7.36
2 and 3	Common-Emitter	7.40	8.90	6.70	2.20

These values align with the designed operating points, validating operation. The transistor's DC node points are stable and expected to show how a superimposed signal will have a full swing output, far from clipping at the rails. The collector-emitter voltages are also sufficient to prevent saturation-caused clipping.

Following this, the quiescent currents can be simulated to further verify designed operating points. The current at the same nodes were plotted below in Figure 8.



(a) Quiescent Currents — Stages 1 and 4 (Common-Collector)



(b) Quiescent Currents — Stages 2 and 3 (Common-Emitter)

Figure 8: Simulated quiescent currents at key nodes for each amplifier topology. Stages 1 and 4 share identical common-collector configurations, while Stages 2 and 3 share identical common-emitter configurations; therefore, each pair exhibits identical DC current behavior.

These results further verify the designed values, with the emitter-follower stages have a collector and emitter current of roughly 1.01mA, while the common-emitter stages are held at 1.45mA.

The close alignment between simulated and calculated values validates the bias network design and confirms that each transistor is operating in the linear, forward-active region.

To further analyze the circuits bias stability and robustness, a parametric sweep of transistor DC current gain and early voltage was performed across the datasheet specified values. According to the datasheet (and many small-signal NPN transistors), the β value can alter (with a 1mA input) from 80 to 300. The early voltage can vary from 20V to 300V. These were the values that were analyzed in order to determine if the collector current had limited variation (thus the entire circuit behavior had limited variation) in correlation to changes in the transistor. The collector current for each transistor across these values is represented below in Figure 9.

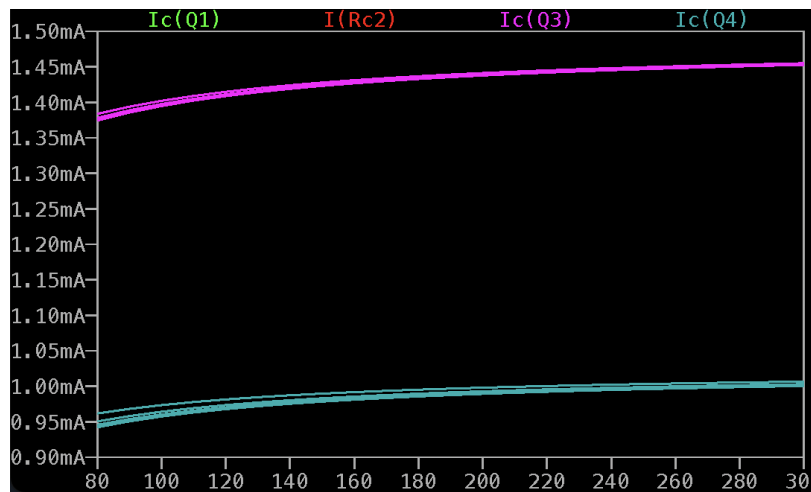


Figure 9: Collector Current changes across β 80 300 and Early Voltage from 20V to 300V.

From this, a quantitative analysis can determine that the percent change in collector current remains

well below the specified allowed amount. This is recorded below in Table 8.

Table 8: Collector Current Variation Under β Sweep (80–300)

Transistor	$I_{C,min}$ (mA)	$I_{C,max}$ (mA)	$I_{C,nom}$ (mA)	% Variation
Q3 (RC2)	1.38	1.45	1.415	4.95%
Q4	0.95	1.00	0.975	5.10%

The small variation in collector current across the wide parametric sweep confirms that the bias network is largely independent of the transistor being used across typically beta and early voltage values. This validates the bias configuration and emitter resistor configuration. The high divider current provides a negative feedback, allowing change in beta to have a minimal influence on quiescent operating points. This parametric sweep confirmed that the design remains stable under worst-case transistor variation, validating component selection and design strategy.

The power dissipated in the transistor can then be determined through the power equation shown below in Equation 29.

$$P \approx V_{CE}I_C : (1.01mA + 1.41mA + 1.41mA + 1.01mA)15V = 72.6mW \quad (29)$$

This is very low power consumption, conservatively below the required specification, and very good for an analog amplifier.

Although each stage was analyzed independently in this section, interstage loading is accounted for in small signal calculations and confirmed through Spice simulations to ensure that overall behavior and meeting of requirements is not obscured from non-idealities.

Transient Analysis and Frequency Response

Following the steady state analysis that worked to characterize the idle function of the circuit, a transient simulation was performed to gain insight into how input variations affect dynamic system behavior.

To begin, the same circuit as shown in Figure 6 was used with a sinusoidal input with a frequency content within the intended pass-band and a small amplitude. The exact parameters used as the input voltage are outlined below in Table 9.

Table 9: Input Signal Parameters for Transient Analysis

Parameter	Value	Units
DC Offset	0	V
Amplitude	5	mV
Frequency	10	kHz

The input and output response was then be simulated to determine if there is a clean output with a minimum 5V peak-to-peak swing, this is shown below in Figure 10 showing the overall output (left) and a zoom in on the input (right).

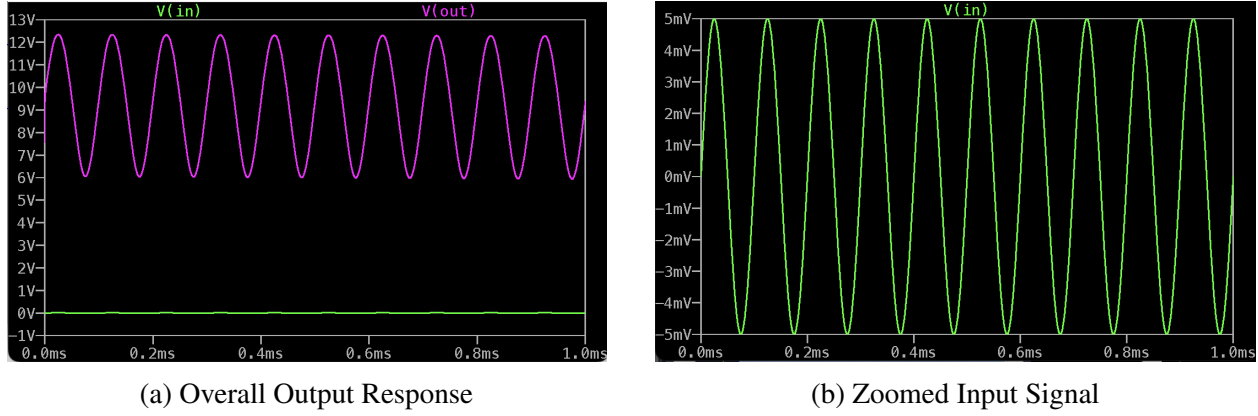


Figure 10: Transient simulation results using a 5 mV, 10 kHz sinusoidal input. The output waveform (left) demonstrates clean amplification with a minimum 5 V peak-to-peak swing, while the input waveform (right) shows the small-signal excitation applied to the amplifier.

It can be qualitatively observed that the output signal has preserved the original waveform free of clipping and distortion. Alongside this, when input 10mVpp, the output swing is greater than the 5Vpp requirement.

Given that the input signal is 10mVpp, and the output is 6.3Vpp, the simulated voltage gain can be determined to be 630, shown below in Equation 30.

$$A_v = \frac{V_{outpp}}{V_{inpp}} : A_v = \frac{6.3}{0.01} = 630 \quad (30)$$

This can also be viewed in dB, shown below in Equation 31.

$$A_v(dB) = 10\log_{10}(630) = 56dB \quad (31)$$

This result is consistent with the analytically predicted gain with minor deviation attributed to transistor parameter variation, and loading effects between stages. The large amplification confirms that the cascaded gain stages function as intended with a gain significantly above the engineering required value.

An experimental confirmation of the gain and output swing was then analyzed below in Figure 11.

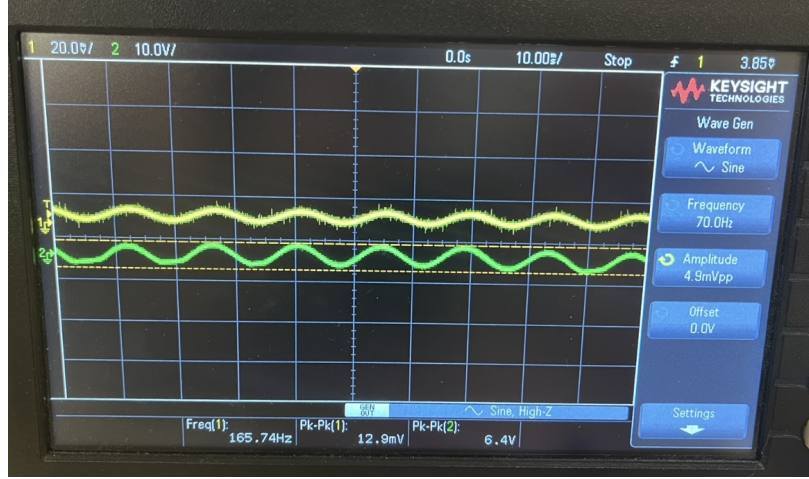


Figure 11: Experimental Results of Input Signal and Amplifier Output Signal

It can be seen that the input voltage (yellow trace) has a peak to peak voltage of 12.9mV and a frequency of 166Hz. This signal was then amplified to 6.4V peak to peak as shown in the green trace.

The experimental gain was then calculated, using Equation 30, to be 497 V/V. It can also be qualitatively observed that the output swing shows linear operation without clipping of at least 6.4Vpp. This verifies the analytical findings, confirming real-world use of the amplifier, still aligning with the engineering requirements of gain and output voltage swing.

Following this, a Hybrid-pi small signal model was created as a linear, simplified circuit representation to analyze the transient response. This model also includes the addition of parasitic capacitance and shows the Miller effect.

At high frequencies, the transistors parasitic capacitances begin to limit performance, lowering bandwidth. The two dominant parasitic capacitances in the BJT are the base-emitter capacitance C_π and the base-collector capacitance C_μ , which can be approximated from the 2N3904 datasheet to be 18pF and 4pF respectively. Because C_μ connects the output and input nodes, it is subject to the miller effect, where the input capacitance is shown below in Equation 32 below.

$$C_\pi + C_\mu(1 - A_v) \quad (32)$$

For the common-emitter stages, per-gain magnitude was roughly 34.4V/V, miller multiplication increases the input capacitance to about 150pF. This forms a low-pass pole, reducing the upper cutoff frequency and contributing to phase lag. The emitter-follower stages, however, have a gain near unity, resulting in minimal impact from the Miller effect. These values were calculated for each stage within the design section.

To mitigate bandwidth reduction, gain was distributed across two stages rather than one singular stage. This limits Miller amplification which is a function of voltage gain. This ensures the amplifier is able to satisfy the upper frequency cutoff within the specified requirements. Figure 12 below shows the created simulation with the parasitics.

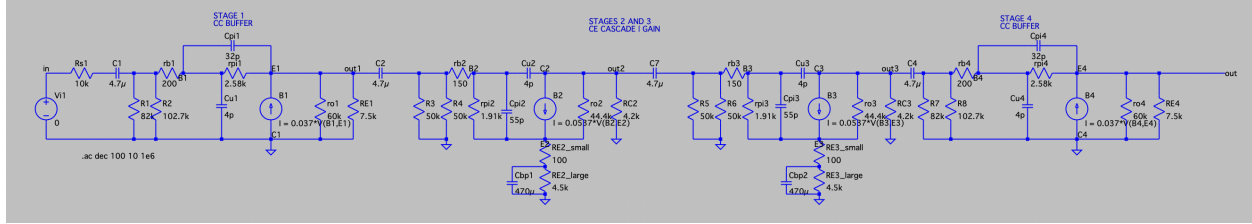


Figure 12: LTSpice Circuit Schematic of Hybrid-Pi Small-Signal Model

In this, the input signal is set to 0V DC and 1V amplitude AC. A sweep across frequencies was then performed to analyze the frequency response, as shown in the Bode plot below in Figure 13.

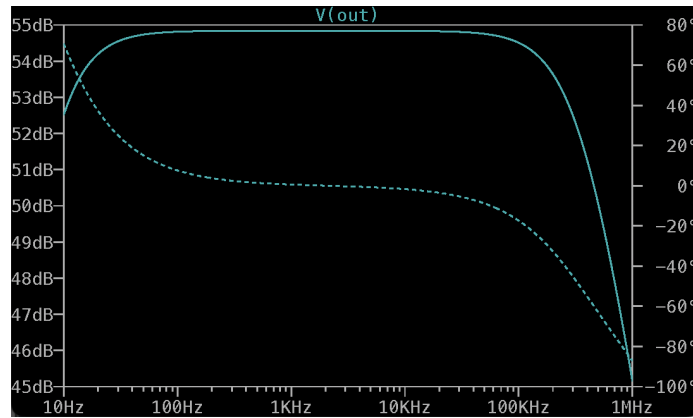


Figure 13: Bode Plot showing Frequency, Phase, and Gain of Amplifier

This plot shows that the mid-band gain in the flat region is held at roughly 55dB, validating the DC simulated gain. The low frequency cutoff point (3dB point) is held at less than 10Hz, while the higher frequency cutoff (3dB point) is held at 344.4kHz. This verifies that the pass-band exceeds the required values of 100Hz - 200kHz. Below the low frequency cutoff, the slope is approximately +20dB/decade, indicating the presence of a dominant single pole at low frequencies, confirming that capacitor choices set the lower cutoff. The higher frequency cutoff appears to have a greater slope magnitude caused by transistor parasitic capacitance, the miller effect, and junction capacitance being the limiting factor.

The phase makes sense and is expected given the circuit topology. The two emitter stages each contribute a 180 degree inversion, so effectively there is a net 0 degree inversion, shown in the flat middle sectioned line on the bode plot. At low frequencies the coupling and bypass capacitors create high-pass poles that produce phase lead until the pass-band is reached. At high frequencies, the phase becomes increasingly negative as transistor parasitics introduce additional poles.

The bode diagram was then plotted from 10Hz to 1MHz on an oscilloscope to experimentally confirm the bandwidth, as shown below in Figure 14.

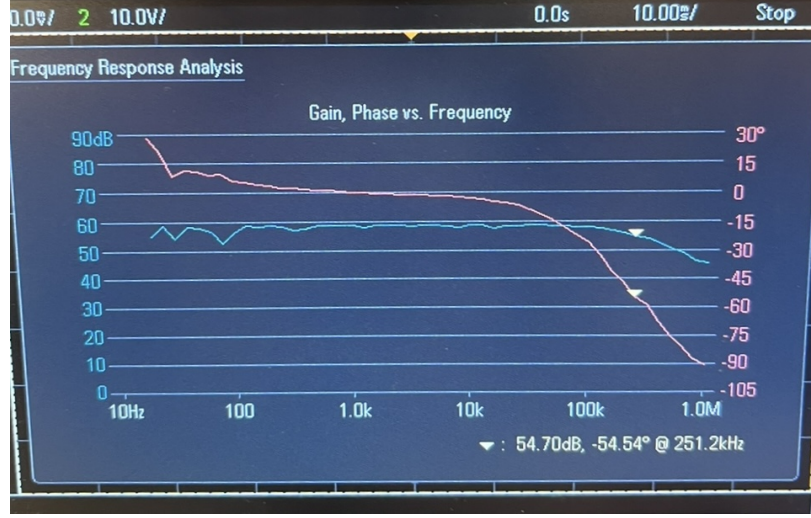


Figure 14: Caption

From this, the experimental bandwidth is determined to be from 94Hz to 251.2kHz. This confirms frequency response in compliance with the bandwidth engineering requirement of minimum 100Hz to 200kHz.

Constraints, Limitations, and Issues

The final amplifier design satisfies all specified performance requirements, however, achieving these specifications required consideration of engineering device tradeoffs, practical limitations, discrepancies across validation techniques, and a deep understanding of constraints and sensitivities. The following discussion outlines these tradeoffs and practical issues in details and explores future remediation to create a stronger, more robust circuit.

Tradeoffs

The amplifiers design required balancing of many competing performance objectives such as voltage gain, bandwidth, bias stability, power consumption, and output swing. Because improving one parameter would often degrade another, component values were meticulously tested and chosen to satisfy all specified requirements.

5.1.1 Gain vs. Bandwidth

Referring back to the original schematic, the gain is primarily set by the collector resistors, the value of the un-bypassed resistor (R_{Esmall}), and the small signal emitter resistance (r_e). The relationship is represented numerically below in Equation 33.

$$A_V \approx \frac{-R_c}{r_e + R_{Esmall}} \quad (33)$$

Bandwidth is controlled by transistor parasitic capacitance, the load resistance seen by the collector, and the collector resistor. The equation for miller capacitance is shown below in Equation

34.

$$C_M = C_\mu(1 - A_v) \quad (34)$$

Due to this, when gain increases, miller capacitance decreases, causing a decrease in the upper cutoff frequency. The resistor collector value is also inversely related to the high frequency cutoff, meaning that increasing this value increases gain, but decreases bandwidth. In the final design, the total gain was distributed across two common-emitter stages to limit the effects of the miller effect, preserving an upper cutoff frequency while allowing sufficient gain.

Stability vs. Gain

Introducing a non-bypassed (in the mid-band) emitter resistor provides a negative feedback to improve linearity, bias stability, and reduces sensitivity to transistor variation. However, this emitter degeneration reduces voltage gain significantly. Without an emitter resistor in a common-emitter topology, the gain is the quotient of the collector resistor and the small-signal emitter resistance, shown below in Equation 35.

$$A_v \approx \frac{-R_c}{r_e} \quad (35)$$

Adding the emitter resistance makes the formula into that shown in Equation 33. This design has an un-bypassed emitter resistor of 100Ω , meaning that the addition of this resistor reduces gain by a factor of 9. Even though this is a large gain reduction, it makes the circuit much more stable and linear, making behavior predictable and stable to variability. The lowered gain was counteracted by cascading two common-emitters together to increase gain while ensuring stability.

5.2.1 Bias Current vs Power

Another design tradeoff was between the selection of quiescent bias current and power consumption. Increasing collector current improves the amplifiers performance by increasing transconductance and reducing intrinsic emitter resistance. This can lead to an increase in possible gain and improve the drive capability. However, increasing the bias current will increase power consumption and reduce efficiency. In this design, the power can be found as the summation of the product of all transistor quiescent collector current and the power supply. Due to this, the buffer stages bias near 1.01mA , while the gain stages bias at 1.41mA . With a single 15V supply power, the quiescent power consumption of this circuit was determined to be 72.6mW , remaining below the 150mW requirement while providing sufficient transconductance for mid-band gain and bandwidth.

Gain Discrepancy

The largest discrepancy occurred between the numerically, simulated, and experimental value for the total gain. The numerically determined gain, shown in Equation 28 is determined to be 776V/V . However, this value was determined to be 560 in both the DC and small signal simulation and only 497V/V in the experimental Bode plot. This is because the numerically determined value assumes each stage behaves completely independent, with gain computed in isolation. Within the real cascade, there are several inter-stage complications that lowers the true gain value. This reduction is expected as inter-stage loading, early effect in output resistance, and loading-dependent gain effectively lowers the total gain. Alongside this, stray capacitances within the breadboard and

wires will cause Miller effect amplification and thus lower total gain/bandwidth. These non-ideal interactions lower effective per-stage gain in the assembled amplifier. However, this was accounted for within the design, making the ideal numeric gain significantly higher than the required value, allowing for a small drop due to non-idealities without dropped below the specification. Due to this, all values computed or determined for the voltage gain surpass the required value.

Practical Testing Issues

This design is required to allow for a 5V peak to peak output swing. As the gain was calculated to be roughly 560, this means that an input of merely 9mVpp is required. The first experimental issue came from implementing this. The function generators used in lab were only able to generate 20mVpp which would ideally produce an 11.2V peak to peak output. However, the amplifier is not designed to handle this large of an output swing, leading to clipping on the output. Alongside this, at such low voltage inputs, there is extreme noise distortion that led to illegible results.

To combat this, a thevenin voltage and resistance was created through a voltage divider at the entry of the circuit to attenuate a large, cleaner signal down to the required input magnitude while maintaining the specified 10k Ω source resistance. The new experimental circuit schematic input is shown below in Figure 15 (left) with the thevenin input the circuit is seeing (right), demonstrating the step down of actual function generator input, and input into the amplifier.

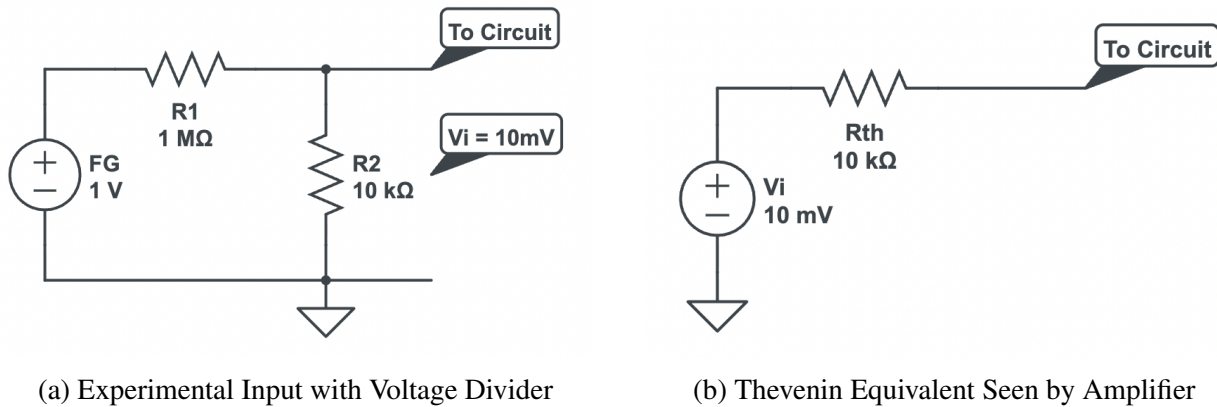


Figure 15: Modified experimental input configuration. The function generator output is attenuated using a resistive voltage divider (left), producing a Thevenin-equivalent source (right) that provides the required millivolt-level excitation and approximately 10 k Ω source resistance at the amplifier input.

Given that R1 would be effectively parallel with R2, and R1 is significantly larger than R2, the thevenin input resistance seen by the circuit is 10k Ω , the required source resistance. From this, it can also be seen that a 1V amplitude function generator input corresponds to a 10mVpp input seen by the circuit, allowing for experimental analyzing of the amplifier without associated low-voltage noise.

Constraints and Sensitivity

The most restrictive design constraint was the requirement that common-emitter stages must satisfy a $g_m R_c < 250$. Since transconductance is a function of collector current, this limit directly couples the allowable collector resistor value to the quiescent bias current. Increasing collector current increases g_m , improving gain potential and reducing r_e . However, with this, $g_m R_c$ increase quickly, violating the constraint. As a result, the design required a balance of current level and collector resistance to achieve the required gain and bandwidth while remaining within this limit.

The circuit also was designed to be fairly insensitive to changes in transistors, accounting for variation in DC current gain (β) and early voltage effects. These influences were prior analyzed within the bias analysis section, and results can be summarized to have a less than 6% variation in collector currents across all beta and V_A values, proving to have low sensitivity to transistor variability.

Further Studies

Possible further work could investigate different circuit topologies that could improve on engineering requirements, For example, exploring the cascode design would significantly reduce the Miller effect and improve bandwidth of the limiting stages. Alongside this, a practical study comparing the BJT and MOSFET could be conducted to evaluate the efficiency of each device when used as a gain or buffer stage, and how they can be coupled together to create a robust, efficient amplifier.

Conclusion

This project successfully designed, created, experimentally validated, and numerically/analytically verified the creation of a multistage bipolar junction transistor amplifier. The design consists of four stages, an input buffer, a cascaded double gain stage, and an output buffer. Evaluation confirmed the systems reliable operation aligning with engineering requirements across possible transistor variability with minimal performance drift. Power consumptions was quantified to be significantly lower than the maximum value, voltage gain was higher than the requirement, output voltage swing was double the required, and bandwidth was larger than the minimum.

Tradeoffs were explored and constraints were identifies including the balancing of voltage gain and bandwidth, physical creation issues, and bias stability gain loss. These limitations were explored, simulated, and used to propose further circuit changes.

Overall, this circuit meets its functional requirements and demonstrates robust amplification under given constraints.

References

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NOTE ON AI USE:

ChatGPT (OpenAI) was used for generating IEEE-formatted citations, assisting in LaTeX table formatting, and rewording selected sentences for improved readability (e.g., prompts such as “What are my grammatical errors”).

Appendix

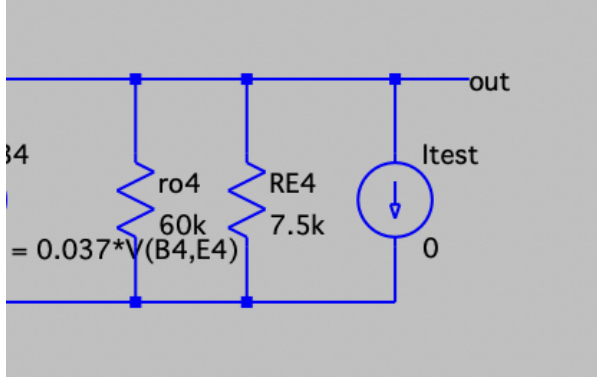
A.)

Test Number	Test Description	Pass/Fail	Notes / Measurements
0100	Prove through LTSpice small-signal AC simulations and supporting hand calculations that the amplifier’s output resistance is less than 50 Ohms while operating within the midband. In simulation, a small test stimulus will be applied at the output, and the resulting impedance will be determined directly. Separately, a hand analysis of the hybrid-pi small-signal model will be performed to analytically compute expected output resistance.	PASS	See Equation 36 and Figure 16
0110	Apply a sinusoidal 100kHz input signal with a 10kOhm series source resistance and adjust amplitude to ensure linear operation. Verify through both oscilloscope measurement and Spice simulation that the amplifier provides a voltage gain greater than 250 within the midband when driven under these conditions.	PASS	See Figures 13, 14, and 11
0120	Sweep the input sine wave frequency across a wide range using a function generator while ensuring a small input amplitude for linear operation (no observable clipping on output). Measure input and output amplitude on the oscilloscope and compute gain magnitude at each frequency, locating the 3dB cut-off frequency points. Verify quantitatively that the lower cutoff is below 100Hz and the upper cutoff is above 200kHz. Confirm through AC simulation that cutoff frequencies relative to midband value reach desired bandwidth.	PASS	Midband from 10Hz to 344.3kHz. See Figures 13 and 14

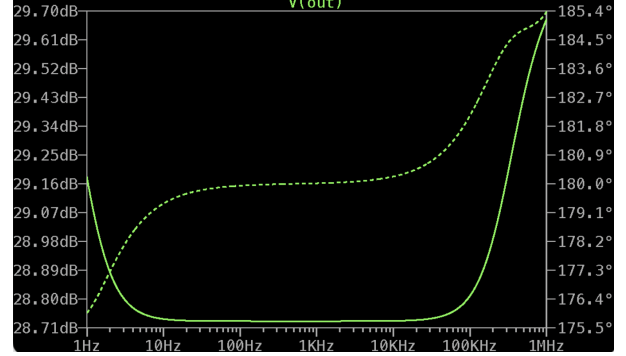
Test Number	Test Description	Pass/Fail	Notes / Measurements
0130	Power the amplifier using a single DC supply set to 15V and verify correct operation, aligning with all working specifications. Measure supply voltage to confirm that it is 15V relative to ground	PASS	See Figure 2
0140	Perform a parametric sweep in Spice to evaluate quiescent collector current sensitivity to DC current gain and early voltage effects. With no input signal, the two variables will be swept across full datasheet defined ranges (Beta = 80–300, $V_A = 20$ –300). An operating point analysis will be performed to compute percent deviation of collector current relative to normal operating point to verify that variation is less than 6% for all parameter combinations	PASS	See Figure ??
0150	Apply a 1kHz sinusoidal input signal via function generator and gradually increase input amplitude until measured output at the overall output node begins to clip. Using an oscilloscope, verify qualitatively that the output waveform clips at a value greater than 5V peak-to-peak.	PASS	See Figures 11 and 10
0160	With no input signal applied, verify total quiescent power consumption using simulation, calculation, and experimental measurement. In Spice, run a DC operating point analysis and record quiescent currents and voltages to compute power. Mathematically determine the total power consumption by summing each transistor's quiescent dissipation. Experimentally, measure the total DC supply current with a digital multimeter to compute quiescent power consumption. Verify that through all three methods, power consumption remains below 150mW under quiescent conditions.	PASS	$P_T = V_{CC} (\sum I_{CQ})$ $15(1.01+1.41+1.41+1.01) = 72.6\text{mW}$ See Figure 8 and Equation 29
0170	Verify qualitatively that the completed amplifier design uses no more than four transistors.	PASS	See Figure 3
0180	[Verifies 1100] Verify mathematically that each common-emitter stage satisfies the hybrid-pi validity constraint by computing the transconductance from the selected quiescent collector current and multiplying the corresponding collector load resistor. Confirm quantitatively that the resulting $g_m R_c$ product is below 250 for each common-emitter stage.	PASS	$G_{m2} = g_{m3} = 0.0537$ $R_{c2} = R_{c3} = 4.2\text{k}$ $g_m R_c = 225.54$

Table 10: Amplifier Verification Test Summary

$$R_{O4} = \frac{R_{\pi}}{1 + \beta} || R_{E4} || r_o : R_{O4} = \frac{2.58k}{201} || 7.5k || 50k = 26\Omega \quad (36)$$



(a) Small-signal model with test current source applied at output to determine R_{out} .



(b) Midband output waveform under small-signal excitation.

Figure 16: Output resistance verification of 26 Ohms and mid-band behavior of amplifier.